NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
THEORY EXAMINATION

Question Paper (Set-1)

Month and year of the Examination: **JAN-2023**

Programme: **B.Tech.** Semester: - **7th Semester**

Subject: - **Operating System (Open Elective)** Course No: -**ITOE-41**

Number of Questions to be Attempted: **5** Maximum Marks: - **50**

Total No. of Questions: **5** Time Allowed: - **3 Hours**

Total No. of Pages used: **2**

Note: - There will be internal choice in Question no. 5

Unless stated otherwise, the symbols have their usual meanings in context with subject. Assume suitably and state, additional data required, if any. The candidates before starting to write the solutions should please check the question paper for any discrepancy and also ensure that have been delivered the question paper of right Course No. and right Subject title.

Ques 1 What is the significance of CPU Scheduling in an Operating system? Suppose that following processes arrive for execution at the times indicated: (10)

Process	Arrival Time	Burst Time(ns)
P1	0.0	8
P2	0.4	4
P3	1.0	1
P4	1.5	6
P5	0.0	5

What is the average turnaround time and average waiting time for these processes with the **Round Robin** Scheduling algorithm? (Quantum time = **1ns**) and **Shortest Remaining Time First** Scheduling algorithm.

Ques 2 Describe Bankers Algorithm in detail. Consider a system with three processes and three resources type and at time T_0 the following snapshot of the system has been taken: (Show all the Steps) (10)

	Allocated			Maximum			Available		
Process	R_1	R_2	R_3	R_1	R_2	R_3	R_1	R_2	R_3
P_1	2	2	3	3	6	8	7	7	10
P_2	2	0	3	4	3	3			
P_3	1	2	4	3	4	4			

(i) Is the current allocation Safe State?

(ii) Would the following requests be granted in current state?

(a) Process P_1 requests (1,1,0)

(b) Process P_2 requests (0,1,0)

Ques 3 (a)	Explain Internal fragmentation and External fragmentation with suitable example.	(4)
Ques 3 (b)	If memory partitions of **200 KB, 400 KB, 600 KB, 500 KB, 300 KB** and **250 KB** are given, how would each of the first fit, best-fit and worst fit algorithm places processes of **357 KB, 210 KB, 468 KB** and **491 KB** (in order)? Which algorithm makes the most efficient use of memory? (Show using diagrams)	(6)
Ques 4 (a)	What is thread, and what are the various types of thread?	(4)
Ques 4 (b)	What is the most important aspect of the critical section? Using semaphore, state the reader's writer's problem and provide a solution.	(6)
Ques 5	The head of moving disk with 100 tracks numbered 0 to 99 is currently serving a request track 50 and moving in left direction. Consider an ordered disk queue with requests involving tracks numbers 45, 21, 67, 90, 4, 50, 89, 52, 61, 87, 25. Find the number of head movements in cylinders using (a) FCFS Scheduling (b) SSTF Scheduling (c) SCAN Scheduling (d) LOOK Scheduling (e) C-SCAN Scheduling	(2*5=10)
	OR	
Ques 5 (a)	Name the different file allocation methods. Explain the linked allocation of file implementation with merits and demerits.	(5)
Ques 5 (b)	Write down different Directory Structure in detail with their advantage and disadvantage.	(5)